

Geometric Memory Gating: Leveraging Fisher Information to Improve Long-Term Credit Assignment in Recurrent Reinforcement Learning

Student name:

Jakub Michałowski

Email:

j.michalowski@tilburguniversity.edu

Student number:

u253954

Supervisor:

Giacomo Spigler

Tilburg University
School of Humanities and Digital Sciences
Department of Cognitive Science and Artificial Intelligence
Tilburg, The Netherlands

February 2026

1 Project Definition

The objective of this project is to implement and evaluate a specific modification to the Long Short-Term Memory (LSTM) architecture within a Reinforcement Learning framework. Specifically, I will introduce an amendment to the forget gate mechanism that incorporates the Fisher Information Metric.

In a standard LSTM, the forget gate determines which information to discard based on the current input and the previous hidden state. The proposed modification involves calculating the Empirical Fisher Information, which measures the sensitivity of the agent’s policy to its internal hidden states, and using this value as an additional input signal to the gate. This allows the gating mechanism to prioritize the retention of information that has a high influence on the agent’s action probabilities.

The modified architecture will be integrated into a Proximal Policy Optimization (PPO) agent. The evaluation will be conducted using the MiniGrid-Memory environment suite, comparing the performance of this "Fisher-Gated" LSTM (FG-LSTM) against standard recurrent baselines in tasks requiring long-term data retention environments where remembering a single, distant hint is the only way to succeed.

2 Motivation

This problem is worth addressing because current recurrent architectures frequently fail to maintain critical information over long durations, a limitation that hampers the performance of autonomous agents in complex environments. Scientifically, this research explores whether shifting memory management from heuristic-based gating to a mathematically rigorous approach using the Fisher Information Metric can improve the stability and retention of internal representations. This is a step toward making neural networks more "aware" of the informational value of their own hidden states. Societally, any advancement in an architecture’s ability to handle delayed dependencies directly increases its utility in real-world tasks, such as autonomous navigation or predictive monitoring, where success depends on accurately linking past signals to future actions. By refining these foundational mechanisms, we can develop more efficient and reliable models for any application requiring long-horizon reasoning.

3 Research Question

To what extent can Information-Geometric Gating (Fisher-Gating) improve long-term credit assignment in recurrent reinforcement learning agents compared to standard LSTM architectures?

4 Background

The challenge of training agents to solve long-horizon tasks with delayed rewards is well-documented in the reinforcement learning (RL) literature. This project builds upon the intersection of Information Geometry, Recurrent Architectures, and the Credit Assignment Problem.

4.1 The Temporal Credit Assignment Problem

The fundamental difficulty of linking past observations to future rewards is known as the Temporal Credit Assignment Problem (Minsky, 1961). Recent work by Pignatelli (2024) highlights that this issue is the primary bottleneck for scaling RL to real-world tasks. The literature demonstrates that even in simple grid-world environments like MiniGrid (Chevalier-Boisvert et al., 2023), standard agents often fail to associate early-episode "hints" with terminal rewards, a failure that results in stagnant success rates once task complexity increases.

4.2 Information Geometry and the Natural Gradient

The theoretical foundation for measuring "information" within a neural network's parameter space is provided by Information Geometry. Amari (1998) demonstrated that the steepest descent in the space of probability distributions is defined by the Natural Gradient. This metric quantifies how much a change in a model's parameters (or, by extension, its internal states) affects its output distribution, providing a more rigorous measure of "importance" than standard Euclidean distance.

4.3 Elastic Weight Consolidation (EWC) and Importance

The use of the FIM to manage "forgetting" was popularized by Kirkpatrick et al. (2017) through Elastic Weight Consolidation (EWC). EWC protects weights that are crucial for previous tasks by slowing down their learning rate based on their Fisher Information value. While EWC focuses on long-term weight retention (Continual Learning), this project proposes an adaptation of this concept: using the FIM to regulate short-term activation retention within a single episode.

4.4 Geometric Selective Forgetting

Recent studies have begun utilizing the FIM to analyze and mitigate "Primacy Bias"—where early experiences dominate a model's behavior. Falzari (2024) introduced Fisher-Guided Selective Forgetting (FGSF), demonstrating that tracking the "trace" of the FIM allows for a geometric understanding of how a network reorganizes its knowledge. Similarly, Golatkar et al. (2020) utilized the FIM for "Weight Scrubbing," proving that information can be selectively removed from a network's weights. These works establish that the Fisher metric is a viable tool for active information management in deep neural networks.

4.5 Recurrent Architectures and Gating Logic

LSTMs were specifically designed to combat the vanishing gradient problem in long sequences by using learnable gates (Gers et al., 2000). However, as noted in recent benchmarks of recurrent agents in POMDPs, these gates are "geometry-blind"; they rely on learned weights that may not capture the true informational importance of a state. This project fills a gap in the literature by testing whether replacing these learned heuristics with a principled geometric signal derived from the FIM can improve memory stability in environments requiring precise temporal credit assignment.

4.6 Summary

The scientific literature identifies the Temporal Credit Assignment Problem as a primary hurdle in training effective memory-based agents. While the LSTM was originally engineered to bridge these gaps via its gating logic, current implementations rely on learned weights that often fail to recognize the intrinsic value of information over long sequences. Simultaneously, the field of Information Geometry has established the Fisher Information Matrix as a mathematically rigorous metric for measuring the sensitivity and "importance" of a model's internal parameters.

While previous research (such as EWC) has successfully used this metric to protect long-term weight knowledge, there is a distinct gap in applying it to the real-time management of hidden state activations. This project aims to bridge that gap by using the Fisher metric to provide the LSTM forget gate with a geometric signal of importance, testing this new approach in the specialized MiniGrid environment.

5 Design Experiment or Dataset Description

This study will utilize the MiniGrid environment suite (Chevalier-Boisvert et al., 2023), specifically focusing on the MiniGrid-MemoryS7-v0 and MiniGrid-MemoryS13-v0 tasks. These environments are designed to evaluate an agent's ability to solve the temporal credit assignment problem. In each task, the agent receives a visual "hint" (a colored object) at the beginning of the episode. It must then navigate a corridor to reach a final decision point where only the object matching the initial hint provides a reward.

The experiment follows a comparative design between two groups:

1. **Control Group:** A standard Proximal Policy Optimization (PPO) agent using a vanilla LSTM architecture.
2. **Experimental Group:** A PPO agent using the proposed Fisher-Gated LSTM (FG-LSTM), where the forget gate is modulated by the diagonal of the empirical Fisher Information Matrix.

To ensure statistical robustness, both architectures will be evaluated across 5 unique random seeds. Performance will be aggregated using the Interquartile Mean (IQM) and reported with 95% stratified bootstrap confidence intervals, following current best practices in RL benchmarking (Agarwal et al., 2021). Primary metrics include Success Rate (the ratio of successful goal reaches), Episode Return, and Sample Efficiency (the number of environment interactions required to reach a 95% success threshold).

6 Instrumentation and Material

The research is entirely computational and does not require physical instrumentation. The study will be conducted using the following software stack:

- **Python 3.10+:** The core programming language for model development and data analysis.

- **PyTorch:** A deep learning framework used for the implementation of the custom FG-LSTM cell, specifically leveraging its `autograd` engine for automated Fisher Information gradient calculation.
- **Stable-Baselines3 / Gymnasium:** For the implementation of the Recurrent PPO algorithm and the execution of the training loops.
- **Gymnasium-MiniGrid:** The environment suite used for simulation and data generation.
- **NumPy, Pandas, Matplotlib/Seaborn:** For processing training logs and generating performance visualizations.

Computation will be performed on a CUDA-enabled GPU (NVIDIA RTX series or equivalent) to facilitate the real-time, high-frequency calculation of Fisher Information metrics required during each forward pass of the model.

7 Statistics, Algorithms, and Software

The primary algorithm used in this study is Recurrent Proximal Policy Optimization (PPO), an on-policy gradient method. To address the research question, I will compare the proposed Fisher-Gated LSTM (FG-LSTM) against a vanilla LSTM-PPO baseline. The vanilla LSTM serves as the baseline method to determine if the geometric gating amendment provides any significant advantage over standard learned heuristics.

7.1 Evaluation and Metrics:

Results will be evaluated based on the agent’s ability to maximize the expected cumulative reward. Since the MiniGrid environment provides a deterministic reward (+1 for success, 0 for failure), the Success Rate acts as our primary metric for accuracy. Ground-truth labeled data is provided by the environment’s internal logic; if the agent reaches the goal tile matching the initial hint, the episode is labeled as a "Success." I will also measure Sample Efficiency, defined as the number of environment interactions required to reach a stable 95% success threshold.

Statistical Analysis: To ensure robustness, the following analyses will be performed:

- **Descriptive Statistics:** I will report the Interquartile Mean (IQM) and 95% stratified bootstrap confidence intervals across 5 unique random seeds to account for the inherent variance in RL training (Agarwal et al., 2021).
- **Inferential Statistics:** A Mann-Whitney U test (non-parametric) will be applied to the final performance distributions to test the null hypothesis that there is no significant difference between the FG-LSTM and the baseline ($p < 0.05$).
- **Geometric Analysis:** I will perform a qualitative analysis of the Fisher Information Trace. By plotting the magnitude of the Fisher signal against the episode’s timeline, I can verify if the "Geometric Shield" correctly identifies the "hint" phase as high-importance.

The implementation will utilize PyTorch's `autograd` to compute the empirical Fisher Information in real-time.

References

- Agarwal, R., Schwarzer, M., Castro, P. S., Courville, A., & Bellemare, M. G. (2022). Deep reinforcement learning at the edge of the statistical precipice. *arXiv preprint arXiv:2108.13264*. doi: 10.48550/arXiv.2108.13264
- Chevalier-Boisvert, M., Dai, B., Towers, M., de Lazcano, R., Willems, L., Lahlou, S., ... Terry, J. (2023). Minigrid & miniworld: Modular & customizable reinforcement learning environments for goal-oriented tasks. *arXiv preprint arXiv:2306.13831*. doi: 10.48550/arXiv.2306.13831
- Falzari, M., & Sabatelli, M. (2025). Fisher-guided selective forgetting: Mitigating the primacy bias in deep reinforcement learning. *arXiv preprint arXiv:2502.00802*. doi: 10.48550/arXiv.2502.00802
- Golatkar, A., Achille, A., & Soatto, S. (2020). Eternal sunshine of the spotless net: Selective forgetting in deep networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. doi: 10.48550/arXiv.1911.04933
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., ... Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526. doi: 10.1073/pnas.1611835114
- Minsky, M. (1961). Steps toward artificial intelligence. *Proceedings of the IRE*, 49(1), 8–30. doi: 10.1109/JRPROC.1961.287775
- Pignatelli, E., Ferret, J., Geist, M., Mesnard, T., van Hasselt, H., Pietquin, O., & Toni, L. (2024). A survey of temporal credit assignment in deep reinforcement learning. *arXiv preprint arXiv:2312.01072*. doi: 10.48550/arXiv.2312.01072
- Shrestha, R. (2023). Natural gradient methods: Perspectives, efficient-scalable approximations, and analysis. *arXiv preprint arXiv:2303.05473*. doi: 10.48550/arXiv.2303.05473